



KnightRate

COP 4331 Spring 2026

Dr. Yadavally

Large Project – Group 7

April 17, 2026 (12:51 – 1:06)



COP 4331 - Large Project - Group 7

- Group Introduction
 - Phillip Lintereur (Mobile / PM)
 - Leandro Vivares (Database / Mobile)
 - Andrew Kelton (Front-End)
 - Adam Betinsky (Front-End)
 - John-Clement Gallo (API)
 - Joshua Berger (API)



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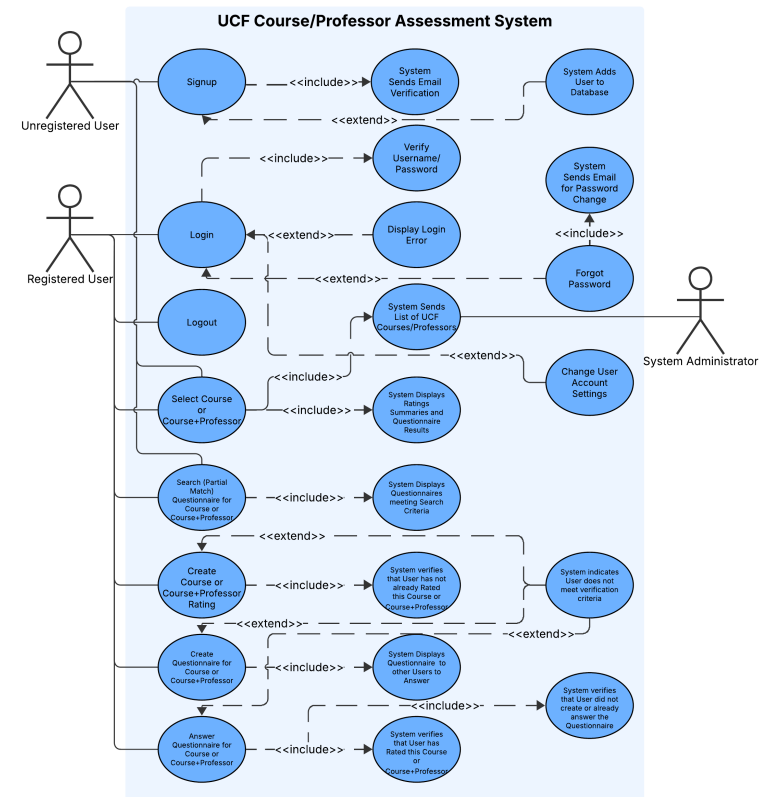
- Large Project Domain Name
 - <http://leandrovivares.com>
- GitHub Repository
 - <https://github.com/AndrewKelton/large-project>
 - Contents
 - Web App / Mobile App Files
 - AI Disclosure
 - Gantt Chart
 - Use-Case Diagram
 - ERD
 - Activity Diagram
 - Prototypes
 - Class Diagrams
 - Unit and Integration Test Results



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• Course/Professor Assessment App

- System
 - Web-App / Mobile-App
- Actors
 - Registered/Unregistered Users
 - System Administrator (Database Manager)
- Use-Cases
 - Select Course or Course+Professor to view Ratings Summary and Questionnaires/results
 - Search (Partial Match) Questionnaires
 - Create Course/Professor Ratings
 - Create and Answer Questionnaires



[illegible]



Item	Description	Role	Start	End	Week 10							Week 11							Week 12							Week 13						
					M 3/23	T 3/24	W 3/25	Th 3/26	F 3/27	Sat 3/28	Sun 3/29	M 3/30	T 3/31	W 4/1	Th 4/2	F 4/3	Sat 4/4	Sun 4/5	M 4/6	T 4/7	W 4/8	Th 4/9	F 4/10	Sat 4/11	Sun 4/12	M 4/13	T 4/14	W 4/15	Th 4/16	F 4/17	Sat 4/18	Sun 4/19
1	Group Intro and Brainstorming Meeting (Week 8)	Group	3/6	3/6																												
2	Database Setup	DB	3/11	3/13																												
3	Feasibility Report	PM	3/11	3/13																												
4	DB Lab Check-In	DB	3/13	3/13																												
5	Group Meeting (Week 9)	Group	3/13	3/13																												
6	Feasibility Report (SUBMITTAL DEADLINE)	Group	3/15	3/15																												
7	Spring Break	Group	3/16	3/22																												
8	API and Front-End Setup	API / FE	3/23	3/27																												
9	Group Internal Check-In (Phone Call)	Group	3/24	3/24																												
10	Professor's In-Class Check-In	Group	3/27	3/27																												
11	API and Front-End Lab Check-In	API / FE	3/27	3/27																												
12	Group Meeting (Week 11)	Group	3/27	3/27																												
13	95% Web App Development (API and Front-End)	API / FE	3/28	4/3																												
14	Mobile App Setup	Mobile	3/28	4/3																												
15	Group Internal Check-In (Phone Call)	Group	3/31	3/31																												
16	Mobile App Lab Check-In	Mobile	4/3	4/3																												
17	95% Web App Completion (Target Internal Deadline)	API / FE	4/3	4/3																												
18	Group Meeting (Week 12) / 95% Web App Demo	Group	4/3	4/3																												
19	100% Web App Development (Front-End)	FE	4/4	4/12																												
20	100% Web App Completion (Target Internal Deadline)	FE	4/13	4/13																												
21	95% Mobile App Development	Mobile	4/4	4/10																												
22	API Documentation	API	4/4	4/10			</																									



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- Things that went well
 - Objectives for app functionality/scope were established early on in project
 - For the most part, stayed on schedule and within scope
 - Review of Pull-Requests and good coordination between all roles
- Things that could have gone better
 - git protocols were inconsistent
 - MERN stack setup was time-expensive
 - Email verification setup (should have used 'Resend' to start)



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- Project/Technology Requirements
 - Web App and Mobile App for Group-Defined Scope
 - MERN Stack (hosed on Digital Ocean Droplet)
 - **M**ongoDB – NoSQL document database (BSON)
 - **E**xpress.js – API Routing, HTTP requests, and responses
 - **R**ead – Frontend JavaScript library for building UIs
 - **N**ode.js - JavaScript runtime environment for servers
 - Functionality
 - Search Function (Partial Match)
 - Email Verification for Registration and Forgot Password ('Resend' was used)
 - Use API's for communication between client and remote server
 - JSON Web Token (JWT)
 - Mobile App
 - Flutter (used AndroidStudio with iOS Simulator)



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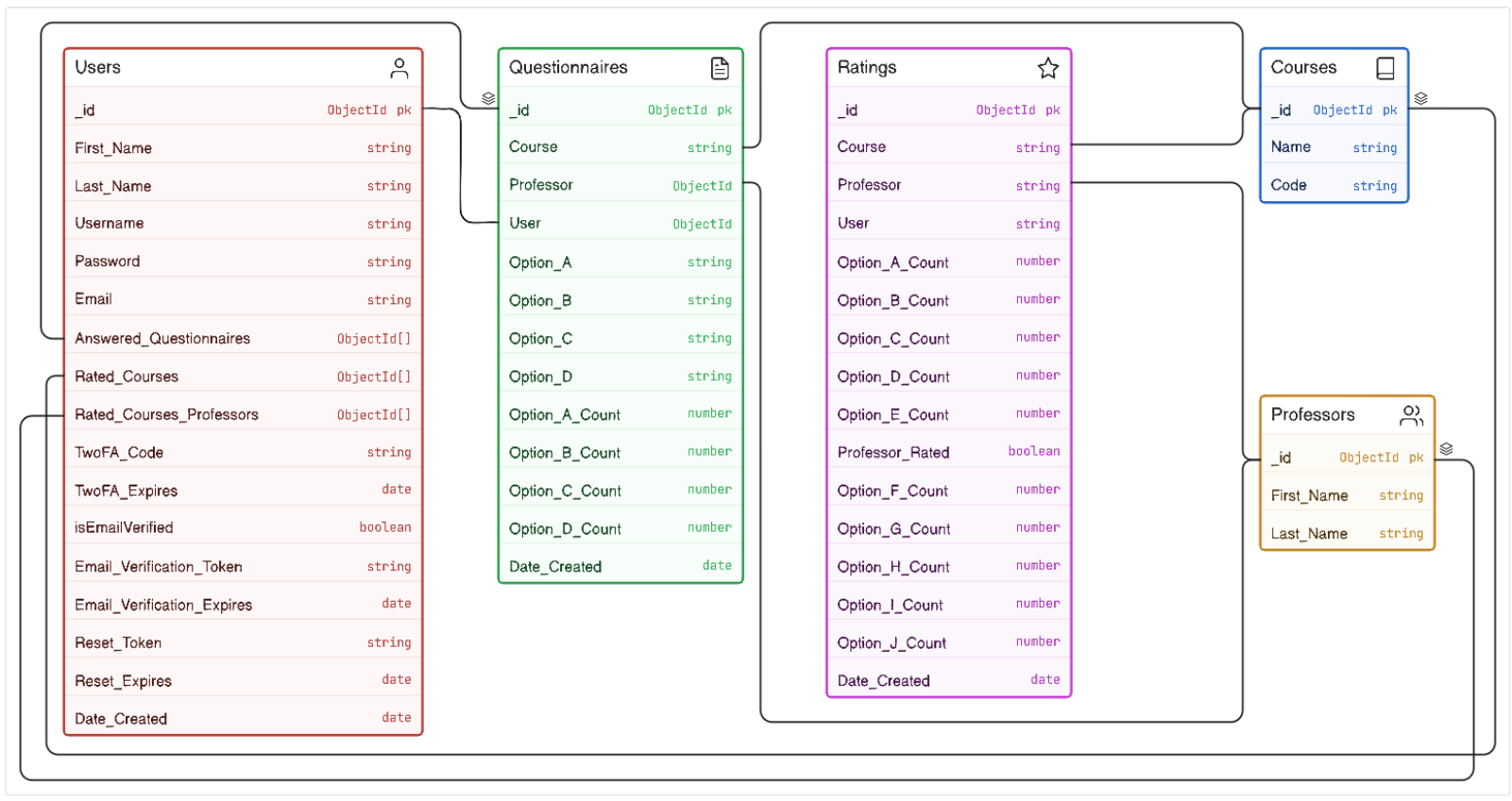
- Database Overview

- Used MongoDB with MongoDB Compass (GUI)
- Used Mongoose in API layer to validate types and insertions
- Relationships between collections were important to establish in the beginning
 - Compared to SQL, relationships were in the API layer rather than the database.
- Design Decisions:
 - Ratings are course / professor agnostic
 - Questionnaires are course / professor agnostic
 - Both must be tied to a user to prevent spam ratings
 - Hashing of password via bcrypt



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ERD





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APIs Created & Used:

API Name	Use	API Name	Use
courses	Get and add courses to database	respondToCO	Respond to CO Q/As
createQuestionnaire	Creates a Q/A from user input	respondToCAP	Respond to CAP Q/As
createRating	Creates ratings from user input	searchCO	Partial search CO Q/As
fetchCO	Fetches course-only Q/As	searchCAP	Partial search CAP Q/As
fetchCAP	Fetches course-and-professor Q/As	twoFactorAuth	2FA functionalities
fetchRatings	Fetches ratings	mailman	Resend functions
login	Login functionalities + verification		
professors	Get and add professors to database		
register	Registers a new user, sends email		
user	Fetches user's data from database		



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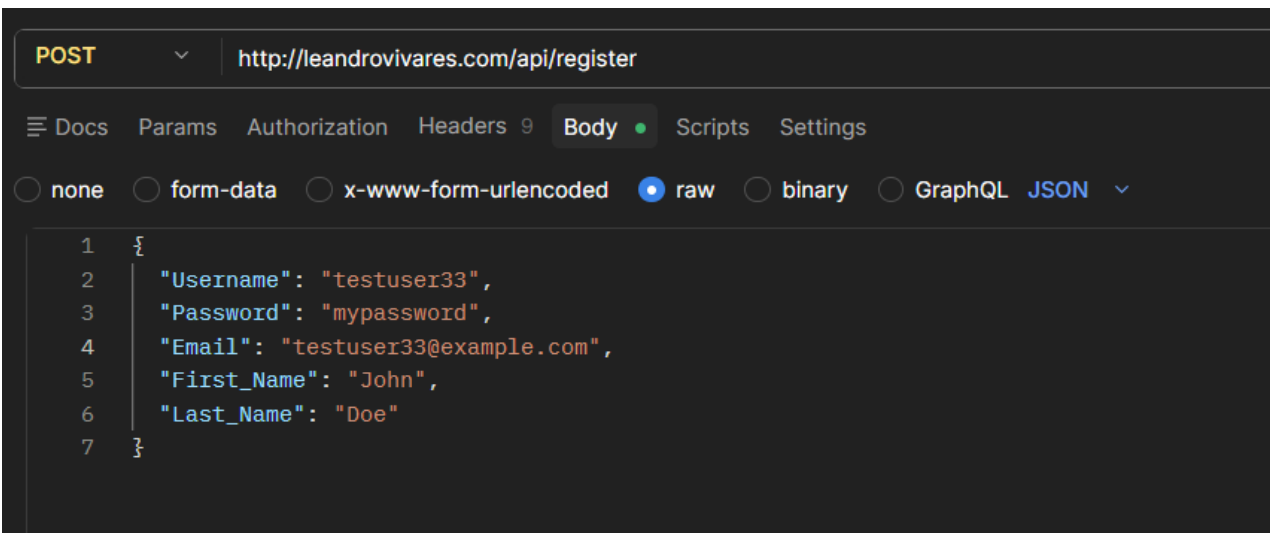
API Documentation Sample for Registration:

A small sample of our API documentation can be seen here.
This documentation is for “register” and can return in three ways.

201 : Account created

400 : User/Email already exist

500 : Server error



```
/api/register:
post:
  summary: Register a new user
  tags: [Authentication]
  requestBody:
    required: true
    content:
      application/json:
        schema:
          type: object
          required: [First_Name, Last_Name, Username, Password, Email]
          properties:
            First_Name:
              type: string
            Last_Name:
              type: string
            Username:
              type: string
            Password:
              type: string
              minLength: 5
            Email:
              type: string
              format: email
  responses:
    '201':
      description: Account created successfully
      content:
        application/json:
          schema:
            type: object
            properties:
              message:
                type: string
              token:
                type: string
              user:
                $ref: '#/components/schemas/User'
    '400':
      description: Validation error or duplicate username/email
    '500':
      description: Server error
```



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Unit & Integration Testing

Backend Testing (Jest)

- 27 unit tests passed: model validation, route logic
- 26 integration tests passed: auth flows, courses & professors, ratings endpoints
- Tools: Jest, Supertest

Frontend Testing (Vitest)

- 11 tests passed: component rendering, auth flows
- Tools: Vitest, React Testing Library

Total: 64 tests, 0 failures

```
Test Suites: 3 passed, 3 total
Tests:       26 passed, 26 total
Snapshots:   0 total
Time:        4.624 s
Ran all test suites matching tests/integration.
```

```
PASS tests/unit/models.test.js
PASS tests/unit/routes.test.js
```

```
Test Suites: 2 passed, 2 total
Tests:       27 passed, 27 total
Snapshots:   0 total
Time:        1.768 s
Ran all test suites matching tests/unit.
```

```
✓ src/tests/components.test.tsx (6 tests) 153ms
✓ src/tests/auth.test.tsx (5 tests) 150ms
```

```
Test Files  2 passed (2)
Tests       11 passed (11)
```



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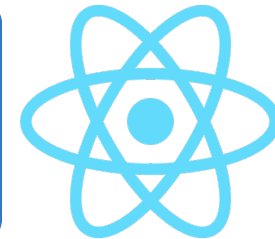
API Demonstration



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- Front-End

- TypeScript – App Logic and Event Handling
- JSX – UI Structure (Buttons, Input Boxes, etc)
- CSS (For Styling)
- Local Development (Node Package Manager for Dependency Management and Live Testing. Communicates with Backend APIs for data and Database access)
- Frontend built using React





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- App Layout
 - System Design
 - Prototypes
 - Functional Specifications

Main Home Page (Desktop)

The screenshot shows the KnightRate desktop home page. The browser address bar displays 'http://leandrovaures.com/'. The page features the KnightRate logo and a 'Welcome to KnightRate' message. Below the welcome message, there are two dropdown menus: 'Choose a Course' (currently showing 'COP 3502C - Computer Science I') and 'Choose a Professor (Optional)' (currently showing 'Tanvir Ahmed'). To the right of these dropdowns, there are two buttons: 'Log Out -> Home Page' and 'User Settings -> Settings Page'. Below the dropdowns, there are two buttons: 'Rate COP 3502C with Tanvir Ahmed -> Create Rating Page' and 'Create Questionnaire for COP 3502C with Tanvir Ahmed -> Create Questionnaire Page'. A red arrow points from the 'Create Questionnaire Page' button to the 'Create Rating Page' button. There are three yellow callout boxes with red headers: 1. 'If no professor is selected, the button will say the course' (pointing to the 'Choose a Professor' dropdown). 2. 'Leave The Professor Dropdown Blank If No Professor' (pointing to the 'Choose a Professor' dropdown). 3. 'If the user is not logged in, the top button says "Login/Sign up" and the Settings button will disappear' (pointing to the 'User Settings' button).



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Course Summary

COP 3502C - Computer Science I

Course Ratings (7 total)

How would you rate this course overall?

★★★★★ 4.50/5

How would you rate the difficulty?

★★★☆☆ 3.00/5

How manageable was the workload?

★★☆☆☆ 1.00/5

Do you feel you will retain the material?

★★★★★ 5.00/5

Would you recommend this course to others?

★★★☆☆ 2.67/5

Course Questionnaire Results

Search Course Questions

How relevant do you think COP 3502C is to industry standards?



Is the class still using C?



Answer this Question

Why do you think this course is paced so quickly?



No answers yet - check back later

Answer this Question

← 1/3 →

Professor Summary

Tanvir Ahmed

Professor Ratings (19 total)

Overall, how would you rate this professor?

★★★★★ 5.00/5

How clearly did the professor explain the material?

★★★★★ 4.88/5

How available was the professor outside of class?

★★★☆☆ 2.00/5

How fairly did the professor grade assignments?

★★☆☆☆ 0.38/5

Would you recommend this professor to others?

★★★★★ 2.48/5

Professor Questionnaire Results

Search Professor Questions

Why are Dr. Ahmed's classes always so late?



Does he still record all of his lectures on Zoom?



Do you think Dr. Ahmed likes his job?



No answers yet - check back later

Answer this Question

← 6/8 →



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KnightRate
http://leandrovivares.com/create-rating

KnightRate
COP 3502C - Computer Science I
Professor: Tanvir Ahmed

Rate

Course Rating Questions
Please rate each item from 1 (Very Poor) to 5 (Excellent)

1. How would you rate this course overall?
Select a rating ▼
2. How would you rate the difficulty?
Select a rating ▼
3. How manageable was the workload?
Select a rating ▼
4. Do you feel you will retain the material?
Select a rating ▼
5. Would you recommend this course to others?
Select a rating ▼

☒ Would you also like to rate a professor?

Professor Rating Questions

Select Professor:
Select a professor ▼

1. Overall, how would you rate this professor?
Select a rating ▼
2. How clearly did the professor explain the material?
Select a rating ▼
3. How available was the professor outside of class?
Select a rating ▼
4. How fairly did the professor grade assignments?
Select a rating ▼
5. Would you recommend this professor to others?
Select a rating ▼

[Submit Rating -> Home Page](#)
[Go Back to Home Page -> Home Page](#)

KnightRate
http://leandrovivares.com/create-questionnaire

KnightRate
COP 3502C - Computer Science I
Professor: Tanvir Ahmed

Questionnaire

Question
What's your question?
Question Input

Answer Choices

Option A
Option A Input

Option B
Option B Input

Option C (optional)
Option C Input

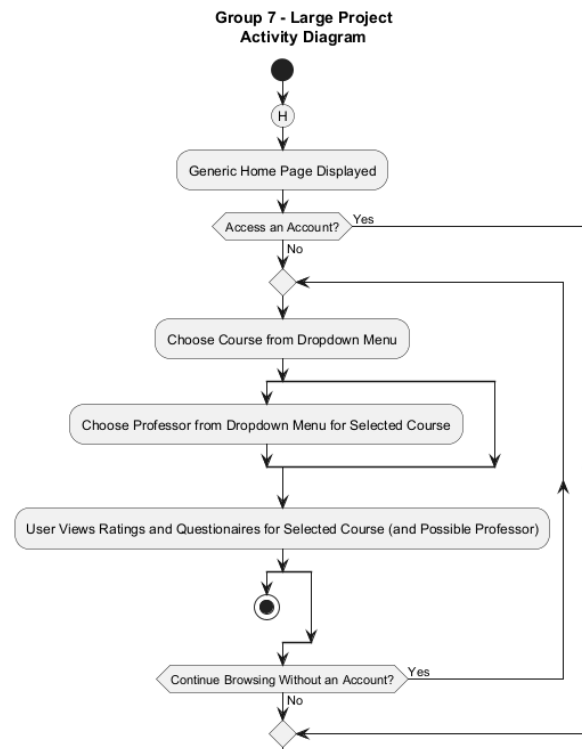
Option D (optional)
Option D Input

[Submit Questionnaire -> Home Page](#)
[Go Back to Home Page -> Home Page](#)



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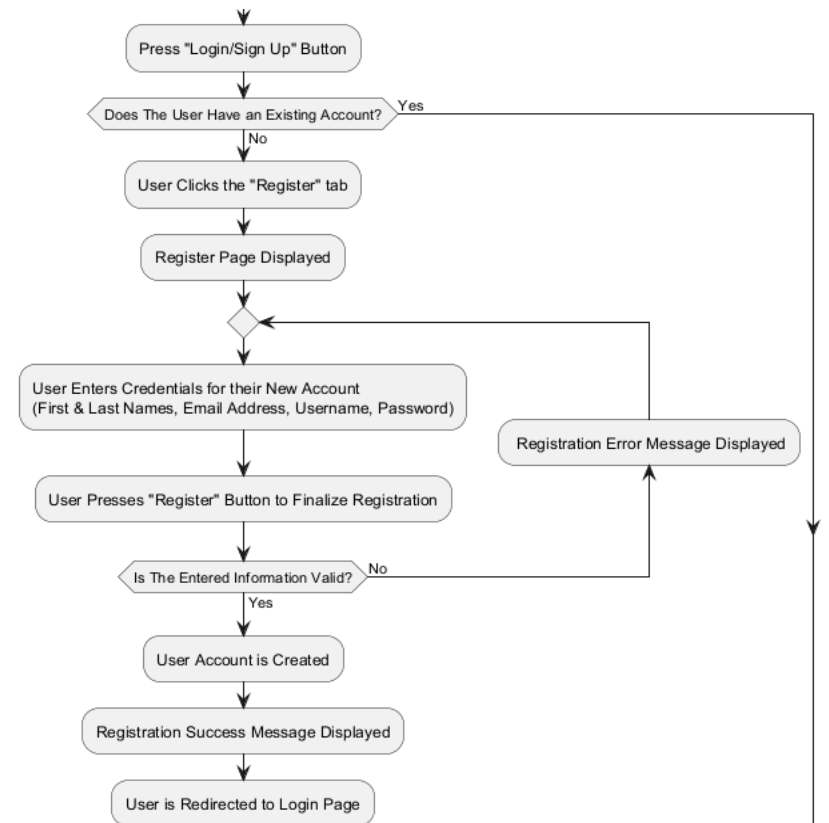
- App Layout
 - System Design
 - Activity Diagram (Part 1A)





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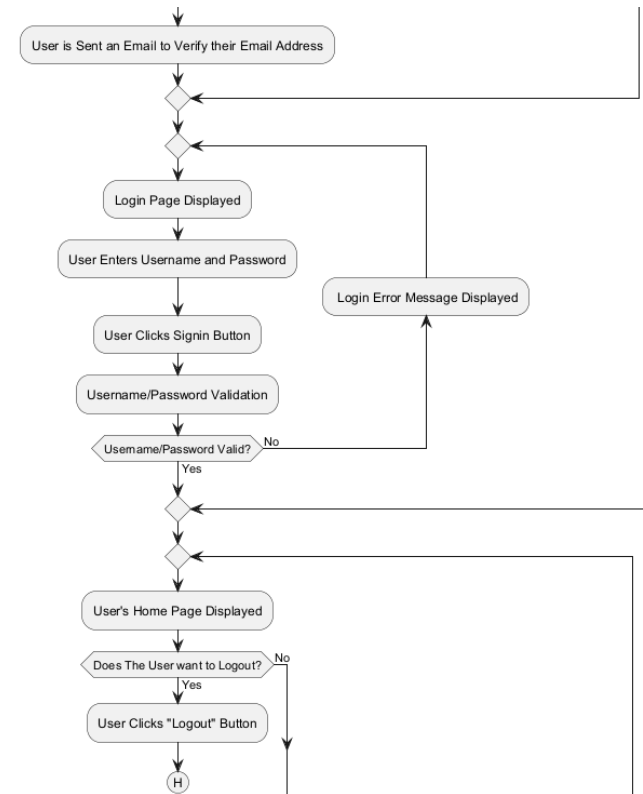
- App Layout
 - System Design
 - Activity Diagram (Part 1B)





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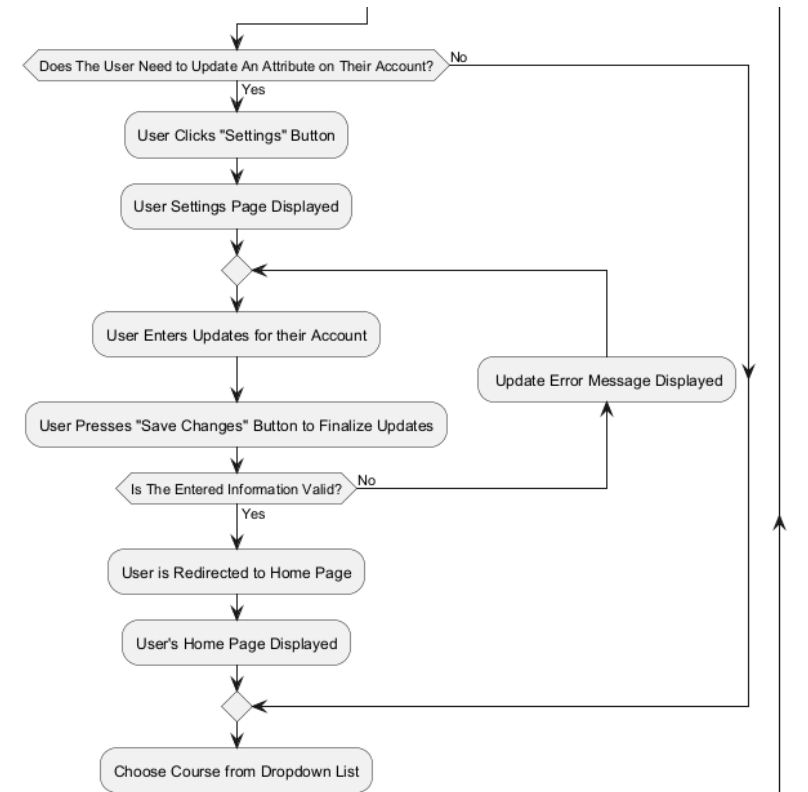
- App Layout
 - System Design
 - Activity Diagram (Part 2A)





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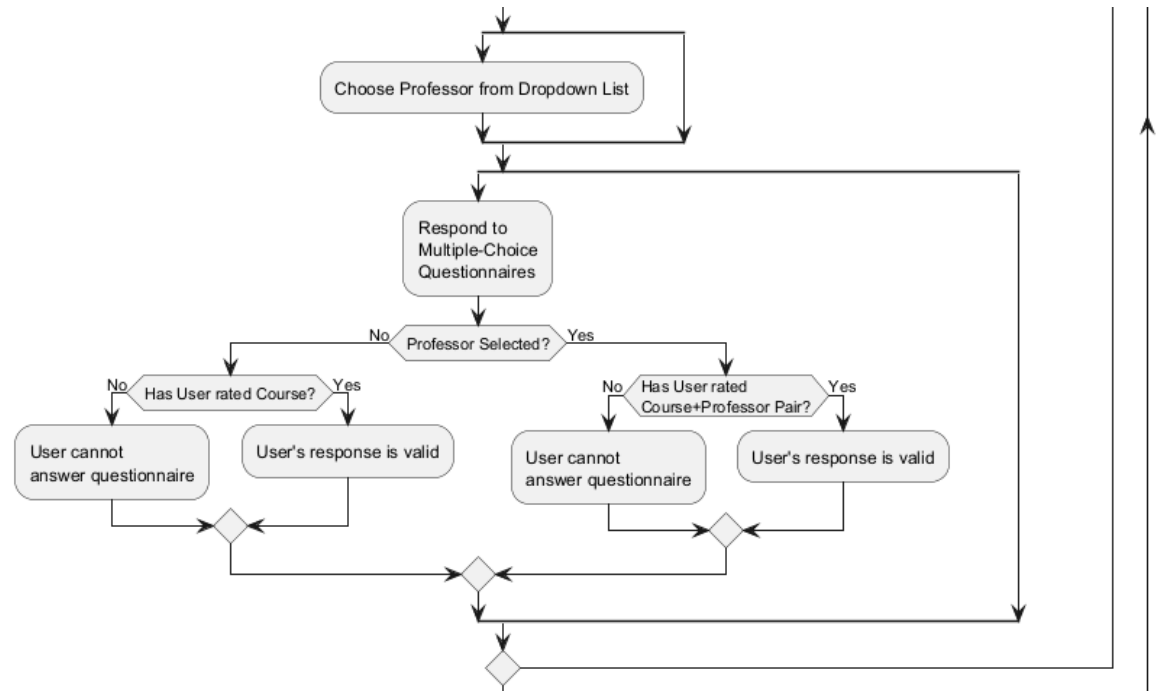
- App Layout
 - System Design
 - Activity Diagram (Part 2B)





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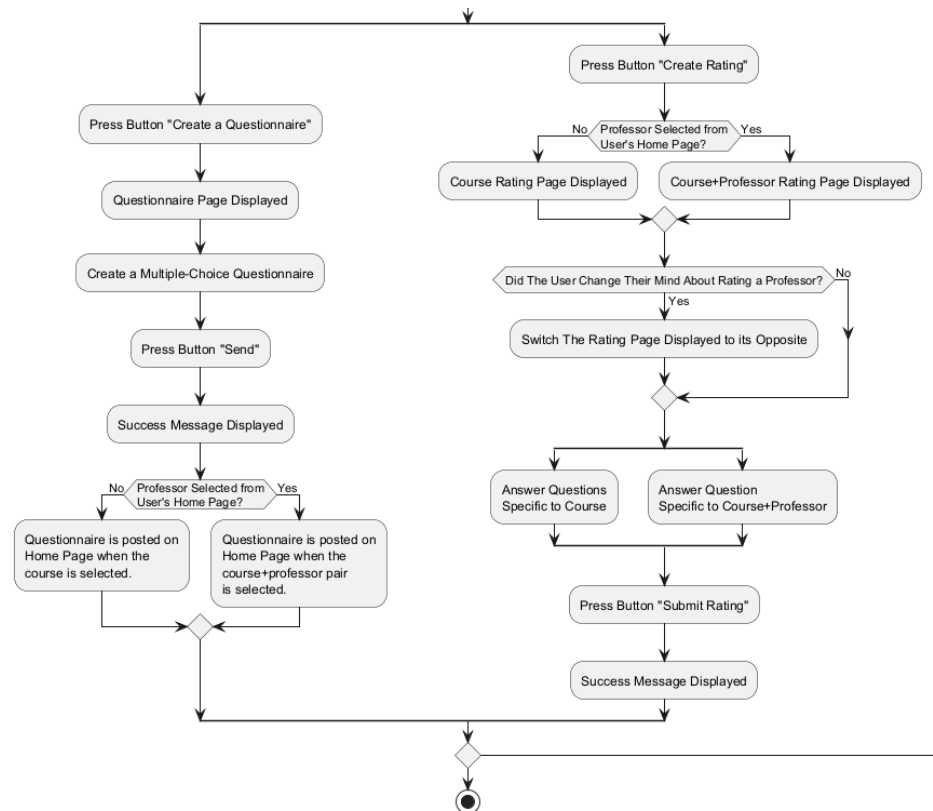
- App Layout
 - System Design
 - Activity Diagram (Part 3A)





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- App Layout
 - System Design
 - Activity Diagram (Part 3B)



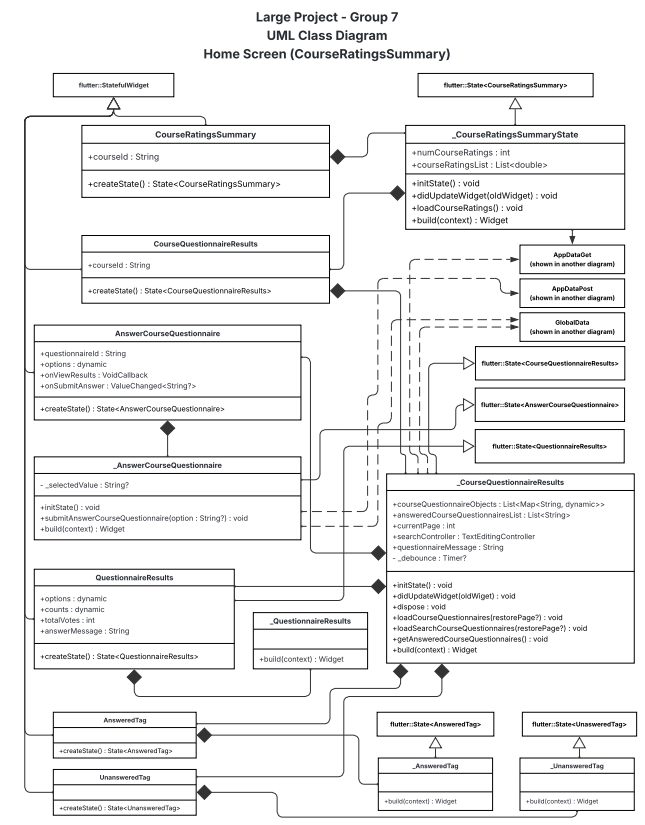
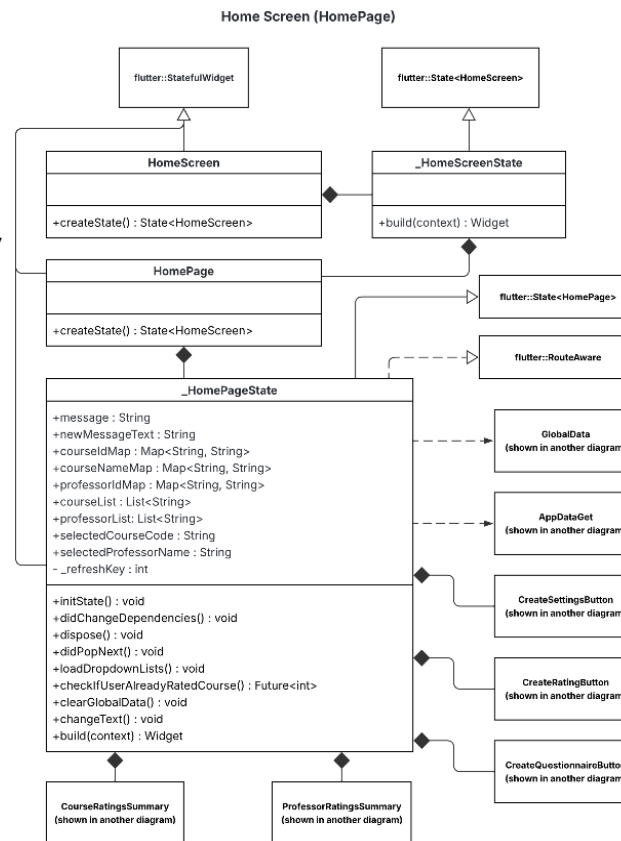


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• Mobile App

Select Class Diagrams

- Home Screen
- CourseRatingsSummary





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Mobile App Demonstration



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Web App Demonstration



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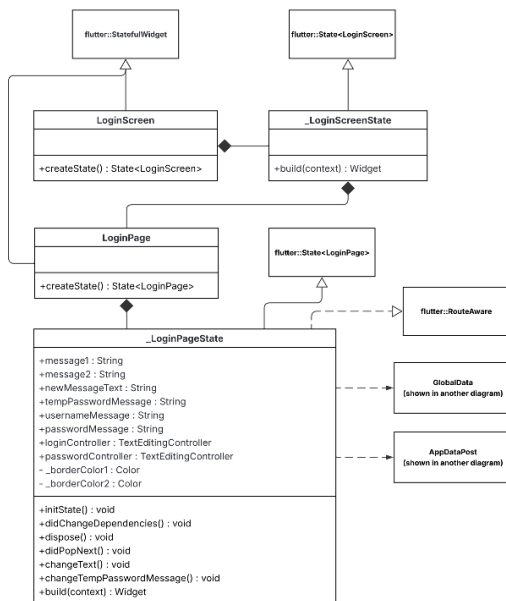
Questions?



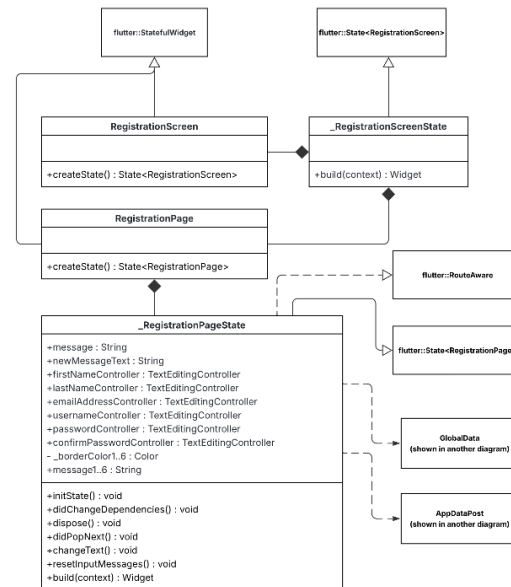
Appendix (Class Diagrams 1)

• Login/Registration Class Diagrams

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UML Class Diagram
Login Screen



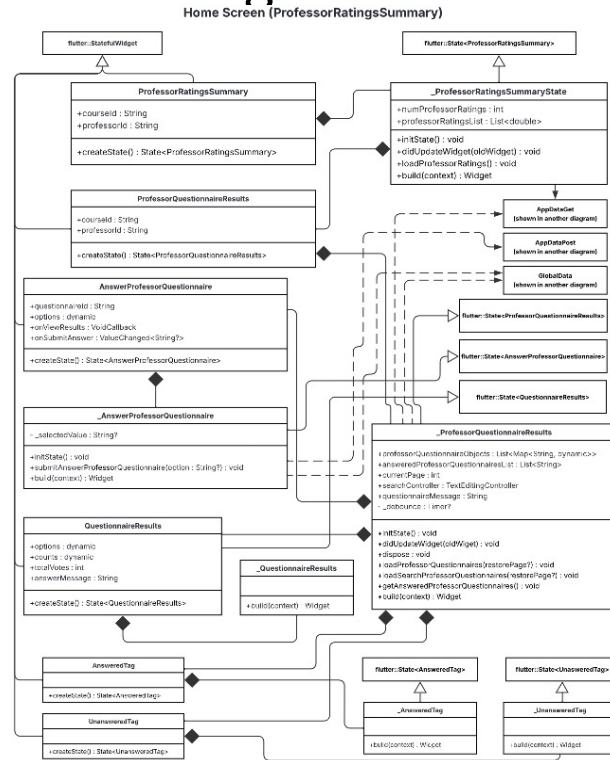
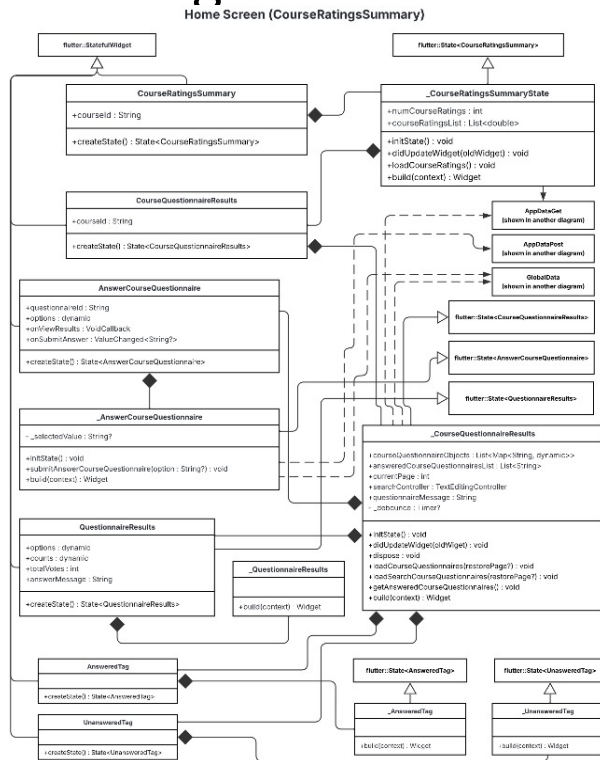
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UML Class Diagram
Registration Screen





Appendix (Class Diagrams 2)

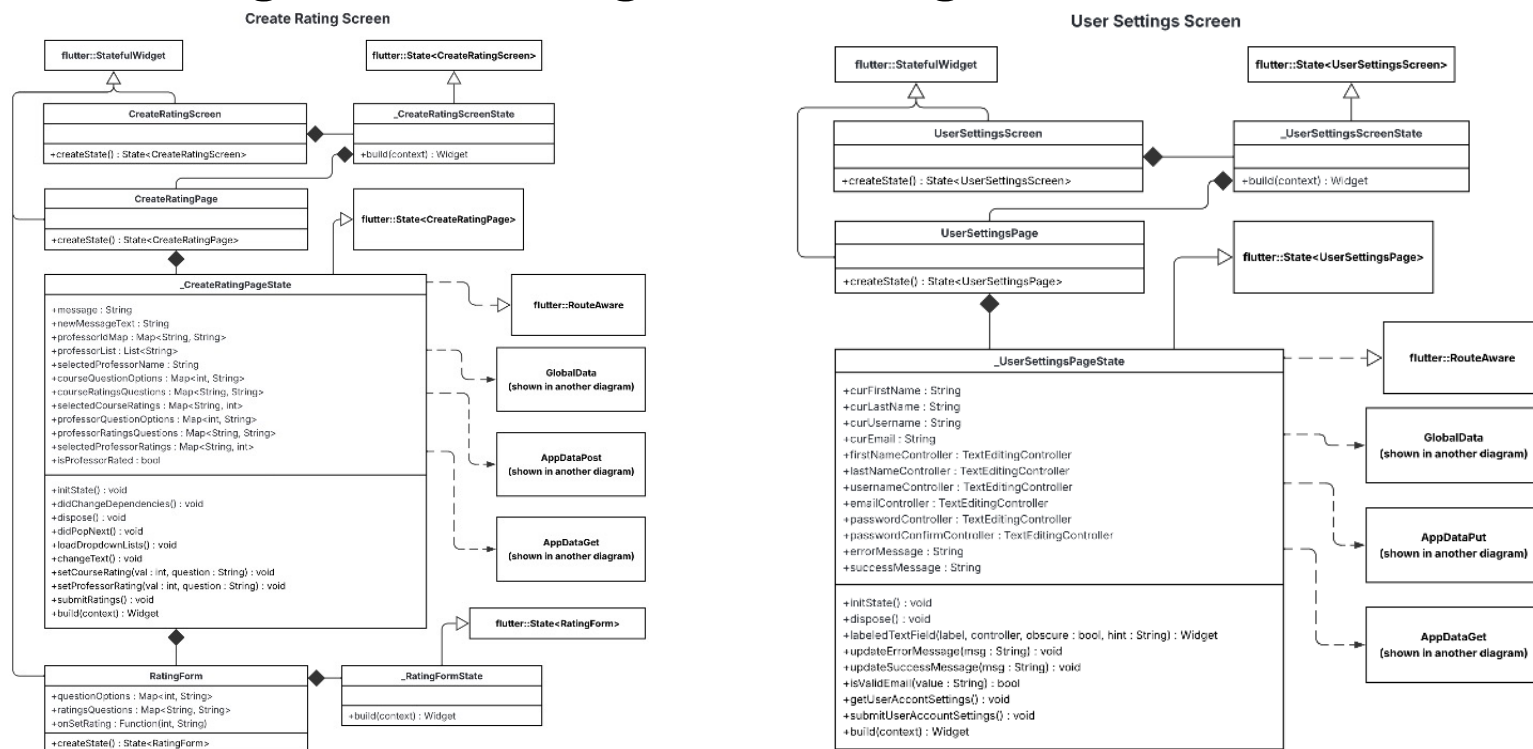
• Course Ratings / Professor Ratings Class Diagrams





Appendix (Class Diagrams 3)

• Create Rating / User Settings Class Diagrams

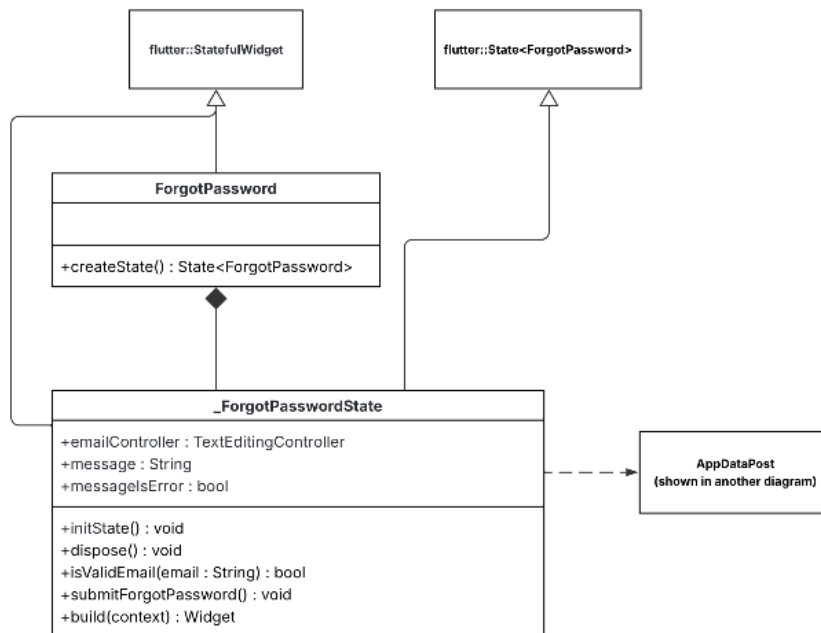




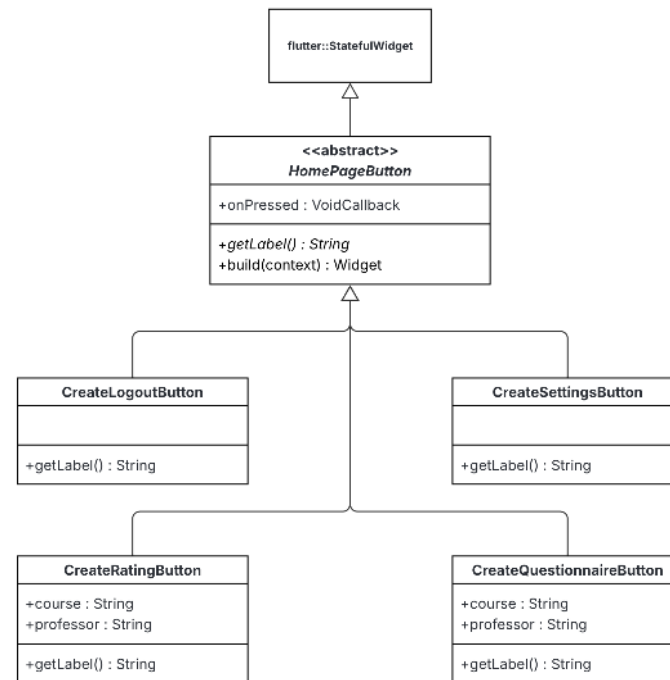
Appendix (Class Diagram 4)

• Forgot Password / Home Screen Class Diagrams

Forgot Password Screen



Home Page Buttons

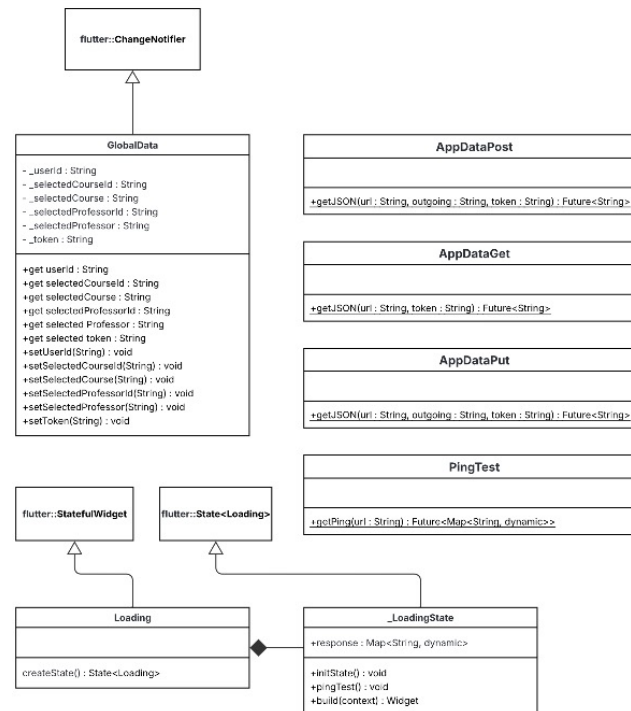




Appendix (Class Diagram 5)

- Miscellaneous Class Diagrams

Global Data, Get/Post/Put API, and Testing Loading





Appendix

- Outside Image Links
 - [TypeScript](#)
 - JSX – UI Structure (Buttons, Input Boxes, etc)
 - [CSS](#) (For Styling)
 - Local Development ([Node Package Manager](#) for Dependency Management and Live Testing. Communicates with Backend APIs for data and Database access)
 - Frontend built using [React](#)
 - AI Disclosure ([Claude and ChatGPT](#))